Ethernet vs Wi-Fi

Detailed Computer Networks Notes • Student Edition

Learning outcomes: After studying these notes, students should be able to define Ethernet and Wi-Fi, compare their operation and performance, and select the appropriate connection for a given situation.

1. Introduction

Ethernet and **Wi-Fi** are technologies used to connect devices to a local area network (LAN). Ethernet commonly uses a physical cable, while Wi-Fi uses radio waves. Both can provide access to local devices and the internet, and both are part of the IEEE 802 family of networking standards.

Ethernet is primarily standardized by **IEEE 802.3**. Wi-Fi is based on the **IEEE 802.11** family. Wi-Fi is a trademark and certification ecosystem managed by the Wi-Fi Alliance; IEEE defines the underlying 802.11 standards.

2. Ethernet: Meaning and Working

Ethernet is a wired networking technology widely used in homes, offices, laboratories, data centres, and campus networks. A device typically connects through an Ethernet network interface and a cable—often twisted-pair copper—to a switch or router. Fibre-optic Ethernet is also common for longer distances and high-capacity links.

Basic working: (1) A network interface sends data as electrical or optical signals over the cable. (2) Ethernet frames carry source and destination MAC addresses and payload data. (3) A switch learns which MAC addresses are reachable through its ports and forwards frames accordingly. (4) A router connects the LAN to other IP networks when needed.

Modern switched, full-duplex Ethernet links allow devices to transmit and receive at the same time. Traditional shared, half-duplex Ethernet used CSMA/CD to detect collisions, but CSMA/CD is not used on modern full-duplex switched links.

3. Wi-Fi: Meaning and Working

Wi-Fi is a wireless local-area networking technology that lets devices communicate using radio waves rather than a dedicated data cable. A wireless access point (AP), often built into a home router, communicates with wireless clients such as phones, laptops, tablets, and smart devices.

Basic working: (1) A device discovers a wireless network and authenticates/associates with the access point. (2) Data is sent in radio frames over a selected channel. (3) The access point bridges wireless traffic to the wired LAN or router. (4) Because radio is a shared medium, devices coordinate access to reduce simultaneous transmissions and interference.

Wi-Fi uses the IEEE 802.11 family. Common generation names include Wi-Fi 4 (802.11n), Wi-Fi 5 (802.11ac), Wi-Fi 6/6E (802.11ax), and Wi-Fi 7 (802.11be). Actual performance depends on the router, client device, frequency band, channel width, distance, interference, and network load.

4. Detailed Comparison

Basis	Ethernet	Wi-Fi
Connection medium	Physical cable, usually twisted-pair copper or fibre.	Radio waves through the air; no data cable to the client.
Main IEEE family	IEEE 802.3	IEEE 802.11
Mobility	Limited by cable length and port location.	Supports movement within wireless coverage, subject to roaming and signal quality.
Typical consistency	Usually stable and predictable on a properly installed link.	Can vary with distance, walls, interference, and other active devices.
Latency	Usually low and consistent on a local switched network.	Can be low, but contention, retransmissions, and signal changes may add variation.
Throughput	Common links include 100 Mb/s, 1 Gb/s, 2.5/5 Gb/s, 10 Gb/s and higher; actual speed depends on equipment and cabling.	Advertised rates vary by Wi-Fi generation and configuration; real throughput is shared and often below theoretical rates.

Basis	Ethernet	Wi-Fi
Interference	Not affected by radio interference; cable quality, damage, and electromagnetic conditions can still matter.	May be affected by neighbouring networks, appliances, obstacles, and other radio devices.
Security	Physical access to a port/cable and network controls matter; wired does not automatically mean secure.	Uses wireless authentication and encryption such as WPA2/WPA3; secure configuration is essential.
Installation	Requires cabling, ports, and often a switch; installation can take more effort.	Usually easier to add wireless devices once an access point is configured.
Cost	May require cable runs, switches, and installation labour; endpoint adapters may be needed.	May reduce endpoint cabling costs, but access points and wireless planning are still required.
Best suited for	Desktops, servers, gaming PCs, workstations, network equipment, and high-demand fixed devices.	Phones, tablets, laptops, visitors, smart-home devices, and locations where cabling is inconvenient.
Shared access	A switched Ethernet port generally provides a dedicated link to the switch, although uplinks and shared network resources can still be bottlenecks.	Devices in the same radio channel share airtime; performance can decline as contention increases.

5. Advantages and Limitations

Ethernet — advantages

• Consistent performance and generally low latency.
• Reliable connection that is less affected by nearby radio activity.
• Supports high-capacity links and is widely used for servers and network infrastructure.

Ethernet — limitations

• Requires physical cabling and available ports.
• Moving a device may require reconnecting or installing another cable.
• Cabling across an existing building can add time and cost.

Wi-Fi — advantages

• Convenient mobility and fewer cables at the user's desk.
• Easy to connect many portable devices and guests.
• Useful in homes, classrooms, public spaces, and places where cable installation is difficult.

Wi-Fi — limitations

• Speed and reliability can change with distance, walls, interference, and congestion.
• Wireless capacity is shared among devices on a channel.
• Requires careful configuration of authentication, encryption, and access-point placement.

6. Which One Should Be Used?

Choose Ethernet when a device is stationary and needs predictable performance—for example, a desktop used for large file transfers, a server, a video-editing workstation, or a competitive gaming PC. **Choose Wi-Fi** when mobility and convenience matter—for example, phones, tablets, laptops, and classroom devices. Many networks use both: Ethernet connects switches and access points, while Wi-Fi connects mobile clients.

7. Common Misconceptions

• **“Ethernet is always faster.”** Not necessarily: a modern Wi-Fi connection can outperform an older or poorly configured Ethernet link. Compare actual equipment and conditions.
• **“Wi-Fi means internet.”** Wi-Fi is a local wireless connection; it can work even when the internet connection is down.
• **“Wired means completely secure.”** Wired networks still need authentication, access control, updates, and monitoring.
• **“Theoretical speed equals real speed.”** Protocol overhead, signal conditions, congestion, hardware, and upstream limits reduce usable throughput.

8. Quick Revision Questions

1. Which IEEE standard family is associated with Ethernet and which with Wi-Fi?

2. Why is Wi-Fi performance more sensitive to walls and interference?

3. What is the role of a switch in an Ethernet LAN?

4. Does Wi-Fi necessarily mean that a device has internet access?

5. Give two examples where Ethernet is a better choice and two where Wi-Fi is more convenient.

Suggested answers: 1. Ethernet: IEEE 802.3; Wi-Fi: IEEE 802.11. 2. It uses radio waves that can be weakened or disrupted. 3. It forwards Ethernet frames between LAN devices using MAC addresses. 4. No; it only indicates a wireless network connection. 5. Ethernet: servers and fixed workstations; Wi-Fi: phones and mobile laptops.

References for further study: IEEE 802.3 Ethernet Working Group (ieee802.org/3), IEEE 802.11 Wireless LAN Working Group (ieee802.org/11), and Wi-Fi Alliance (wi-fi.org). This document is a teaching summary; specific speeds and features depend on the exact standard, hardware, and deployment.